Functional Requirements

ID	Requirement	Module
FR-1	User can upload a food image (JPG, PNG or WEBP)	Food Recognition
FR-2	User can add optional voice (hold to record) or text context	Food Recognition
FR-3	System returns dish name, confidence, serving size (g), calories with range, protein, carbs, fat and a health score	Food Recognition
FR-4	System rejects unsupported file types and oversized files with a clear message	Food Recognition
FR-5	User can submit food for a freshness check and gets a result	Freshness Detector
FR-6	User can submit images or video for hygiene analysis and gets findings	Hygiene Monitor
FR-7	User can ask for cooking or recipe help	Smart Kitchen
FR-8	User enters age, gender, height, weight, activity level, goal, diet type, cuisine, allergies, medical conditions and recent food history	Meal Planner
FR-9	System generates a personalized meal plan that avoids the listed allergies	Meal Planner
FR-10	System shows a readable error if the AI service fails	All
FR-11	Sidebar navigation between modules and an online/offline status indicator	UI

Non-functional requirements

- Response time: image analysis within a few seconds on a normal laptop (TODO: measure and state the number).
- Privacy: health inputs are processed locally.
- Usability: single-page dark UI, works on desktop browsers.